

$$\partial_t u = \kappa \partial_{xx} u, \qquad 0 < x < L, \quad t > 0,$$

$$u(0, t) = u(L, t) = 0, \qquad \kappa > 0,$$

$$u(x, 0) = U_0 \left[\sin \frac{\pi x}{L} + 0.35 \sin \frac{3\pi x}{L} \right],$$

$$\xi = x/L, \qquad \tau = \kappa t/L^2, \qquad v = u/U_0,$$

$$v_\tau = v_{\xi\xi}, \qquad 0 < \xi < 1.$$